TECHNICAL NOTE · OPEN

Hierarchical Bits

A Structural Envelope for Orchestrating Representations — Method, Measurements and Boundaries

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Technical demonstration report — every claim is accompanied by the method and the measured number. 128+ automated tests green; exact correctness as a gate before every measurement.

EXECUTIVE SUMMARY

Hierarchical Bits (BH) began as a hypothesis from December 2025 ("the byte is an implicit tree; hierarchy is interpretation"). This study tested the hypothesis from **nine independent angles**, with real measurements and honest baselines. The result is not "a better codec" — it is the precise delimitation of a paradigm:

```
BH does NOT compress dense signal better than JPEG/WebP/AVIF (it loses — measured).
BH IS a STRUCTURAL ENVELOPE that:
   (a) makes the structure explicit at low cost (0–6% of the file — measured),
   (b) routes each region's residual to the right specialist (orchestration),
   (c) offers multiple reads over the same structure (preview/ROI/proof).
```

Anchor result (measured): a structured document written in the orchestrated format is **2.1× smaller than WebP AND** lets you read any region in **3–55× fewer bytes**, within the same file — something no SOTA tool offers together.

The transversal law (measured across all angles): BH pays off exactly to the extent that the data is **structure** (rule-generated, composite, hierarchical) and not **signal** (noise/perceptual). At the limit, the payload becomes a *program* that generates the data (gains of thousands×); the boundary is not entropy — it is structure recognition.

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1. METHODOLOGY (what makes this study trustworthy)

The entire investigation followed four rules, applied without exception:

1.1 Declare before measuring. Each terrain defined falsifiable claims with a criterion for **success AND failure** BEFORE the measurement. No number was promised a priori.

1.2 Correctness as a gate. Before any performance comparison, exact correctness was verified (bit-for-bit reconstruction, equality of aggregates, valid cryptographic proof). Performance is only measured over a correct result.

1.3 Honest baseline, with the vs-naive / vs-SOTA distinction. Large numbers against naive baselines (full scan, transmitting everything) are marked as such and distinguished from gains against the state of the art (WebP, HNSW, OLAP).

1.4 Real measurement, public self-correction. Real hardware was used (RTX 3060, CUDA events, `nvidia-smi dmon`). When a measurement was biased in favor of the hypothesis, it was corrected in public. Three material self-corrections occurred and are documented (§7) — they are part of the evidence of rigor.

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2. REFRAMING THE QUESTION

The initial question — **"is BH a better codec?"** — was answered, measurement after measurement, with **no**. The right question only emerged at the end of the investigation: **"is BH a structural orchestrator of representations?"** — and the answer is **yes**. The key was a separation that resolves the confusion of half the study:

```
RESIDUAL PROBLEM  ≠  PARADIGM PROBLEM
```

Trying to compress the residual (dense signal) is the problem of specialist codecs — and BH loses at it. The BH paradigm is structure, belonging and reads. Separated, each half has an honest and opposite verdict.

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3. THE TERRAINS TESTED (method + measurement)

3.1 Image codec — BH as a compressor (loses)

Method: quadtree codec with per-node interpretation selection (LEAF/RAMP/DCT), compared against PNG/JPEG/WebP on 4K frames, with matched PSNR.

Measurements: - Matched content (gradient, lossy): BH-ramp **2.4 KB @ 57.9 dB** vs WebP **116.7 KB @ 50.2 dB → 48× smaller, with higher quality**. - Access (thumbnail of a natural 4K photo): BH reads **0.148 MB** vs WebP 0.664 (4×), JPEG 1.0 (7×), PNG 6.5 (44×). - Natural photo compression: **loses** (4–10× larger than WebP). - v0.3 (DCT interpretation): switching the error criterion from $L\infty$ to L2 unlocked the DCT and cut 40–66%, but **did not close the gap** — residual entropy coding is missing. - Heterogeneous image (pure BH-compressor): **loses at all photo fractions (10.75× at 0% photo)** — because text/UI explodes the quadtree and WebP's entropy coding already makes the easy regions cheap.

Terrain conclusion: as a compressor, BH loses. The 48× gain on the gradient was a corner case (perfectly procedural content), not the rule.

3.2 Database — selective read by aggregation

Method: aggregation tree (min/max/sum/count per node) over 1,000,000 rows; metric = rows read; baseline = flat scan.

Measurements:

query	BH reads	flat reads	gain	vs
global SUM	0	1,000,000	∞	naive
SUM range 10%	1,564	99,868	64×	naive
COUNT key 2%	2,048	1,000,000	488×	naive
COUNT correlated value > p99	92,736	1,000,000	11×	naive
wrong axis / independent value / low selectivity	~10^6	10^6	1–2×	boundary

Interpretation materialization: a per-region filter (not aggregated) read everything (1×); with a per-region counter in the node → root read, **0 rows (∞)** — at the cost of storage that scales with cardinality.

Honesty: the gains are enormous vs naive scan, but the mechanism (pre-computed summary) is already state of the art (zone-maps, OLAP, materialized views).

3.3 Verification / Merkle — selective proof

Method: Merkle tree (SHA-256, domain separation) over 1,048,576 items; crypto correctness gate (valid proof verifies, forged proof fails) before measuring; metric = bytes/nodes.

Measurements:

task	BH reads	naive baseline	gain
commitment of 1M items	32 B (1 hash)	33.5 MB	1,048,576×
membership proof	644 B (20 hashes)	33.5 MB	52,103×
locate tampering	40 nodes	1,048,576 re-hashes	26,214×

The proof grows with **log n** (1k → 10 hashes; 1M → 20). Honesty: it is the standard Merkle construction (blockchain/git/CT); the study shows the **unification** with the other terrains, it does not invent crypto.

3.4 Wafer — multiple co-registered layers + scale (film)

Method: quadtree-union over co-registered layers (RGB+depth+ segmentation); per-layer lossless reconstruction gate; measurement of shared-structure vs K independent trees.

Measurements: - Rigid union: co-registered **1.12×**; misaligned **0.32× (loses)**; photo+lum+seg **0.67× (loses)**. - With a derived layer (luminance from RGB): **0.78×**. - With base + local refinement (non-union): **1.12× (wins)** — the natural photo moves from loss to gain; verified by lossless reconstruction. - **Scale proof (film, 720 frames = 30s):** independent 1.00× · temporal (delta between frames) 1.65× · wafer-still 1.12× · **wafer+temporal 2.13×**. The structure-fraction rises from **16.7% → 33.4%** with the temporal — structure only starts to pay off when redundancy empties the payload.

3.5 GPU — data movement (read face, on real silicon)

Method: data-movement measurement (simulation) and real wall time (CUDA events, RTX 3060), with the load visible via `nvidia-smi dmon`.

Measurements: - Simulation (bytes moved, aggregation/LOD/context load): **1,540× less data**. - Light real test (1 GB in VRAM): total reduction flat 2,991 µs vs BH 2.8 µs → **1,087×**; range aggregation **264×**; flat bandwidth **342 GB/s** (pinned at the ceiling of 360 → genuinely bandwidth-bound). - **Heavy test (real load):** 6,000 queries, **3.51 TB scanned**, SM at 100% for 26 s. Diagnosis of two pitfalls: (a) my flat kernel was suboptimal (134 GB/s = 37% of the ceiling), inflating the raw number (4,725×); (b) the BH side at the measurement floor (noise). **Honest triangulated number** (time vs flat-ideal AND data-moved: 3.51 TB vs ~2 GB): **~1,750×**.

Extrapolation (published bandwidth, NOT measured): the batch gain is **hardware-invariant** (~1,755× from the 3060 to the B200 — algorithmic, both scale with bandwidth); the single-query gain **shrinks** on fast cards (992× → 45×), because the kernel-launch floor is fixed.

Native hardware model (labeled projection, not measurement): with the floor at memory-latency level (0.3 µs, physical) instead of launch (2.8 µs, software): single latency **9,747×**, batch **5,702×** (near-memory build), energy **~1,755× less** (∝ data moved — robust, independent of the pJ/B value).

Honesty: the large numbers are vs naive scan; the mechanism (pre-computed aggregate) is standard technique (OLAP/materialized views). The GPU proves that the **read face** is real and fast in hardware; it is not a standalone product.

3.6 AI multimodal storage — where BH was tested away from home

Method: unified substrate (structure + embeddings) vs stitched stack (storage + HNSW + cache + spatial index), over a co-registered asset.

Measurements (storage, slice 1):

d (embedding)	unified vs stitched
8	4.0× (wins)
128	tie
256	0.86× (loses)
768–4096 (real AI)	**loses**

Self-correction (progressive embedding / Matryoshka): by treating the embedding as compressible infrastructure (not irreducible payload) — prefix in the internal node instead of full-d — the verdict **moves**: d=256 → 1.06×, d=1024 → 1.02×, d=4096 → 1.00× (ties). The bulk (leaf embedding) is a Shannon tie; the gain is in the index.

Measurements (access, slice 2): preview 7.6× · aggregate 3.2× · ROI 2.1× · narrow-scoped retrieval 1.4× · **broad retrieval 0.23× (loses)**.

Architectural debt decomposition (real RAG stack): the eliminable debt is **14–49%** (borderline), **dominated by embedding duplication** (dense payload), not by structure. Conclusion: dense AI storage is not BH's terrain.

3.7 Compositional / real density — the strongest signal and its limit

Method (compositional): symbolic data — concepts as equations of shared primitives — represented as an algebraic envelope vs dense vector.

Measurement: 384× smaller (8 MB vs 3.07 GB for 1M concepts), and it answers structural queries ("which concepts use primitive X?") that the dense vector **cannot even formulate**.

Method (real density): intrinsic dimensionality of real embeddings (all-MiniLM-L6-v2 model, 6,000 PT words, PCA).

Measurement: 90% of variance in **160 of 384 dimensions** (~2.4× compressible). **Honesty:** this is *linear* compressibility (already exploited by PCA/PQ/ Matryoshka), **not** symbolic compositionality. Real language data, in embedding geometry, is **not** compositional-symbolic — it is a subspace of medium dimension. The 384× holds for data *that is* composition; how much of the real world is compositional remains the open bet (of the symbolic domain).

3.8 Decode-program — payload becomes program (the deep form of the law)

Method: the header carries the *program* that generates the region (not the result); the decoder executes the program. Test of generality (several families) + noise + hidden cost.

Measurements:

rule-generated family	WebP	program	gain
rings	37.8 KB	13 B	2,904×
waves	46.4 KB	13 B	3,572×
checkerboard	2.4 KB	3 B	809×
gradient	2.1 KB	1 B	2,096×

Under noise (procedural base + noise α): the program-aware approach **keeps the structure advantage** at all noise levels (subtracts the known base; whole-WebP pays for the structure it cannot separate). **Hidden cost (declared):** the encoder needs to **discover** the program — the inverse problem, trivial for a known family, **undecidable in general** (Kolmogorov complexity is uncomputable). BH's boundary is not the residual's entropy — it is **structure recognition**.

3.9 Orchestration + union — the formulation that wins for the right reason

Method (orchestration): BH routes each region to the local specialist (flat→constant, gradient→formula, text→PNG, photo→WebP), compared against one-codec-on-the-whole, on two content types.

Measurements:

content	whole-WebP	orchestrated	result
photo-heavy	10.7 KB	10.8 KB	1.01× (ties)
document (closed forms)	4.8 KB	**2.2 KB**	**2.1× smaller**

Splitting into WebP per region (without routing paradigms) **loses** (1.10×) — the gain comes from cross-paradigm routing, not from the cut.

Method (envelope cost): decomposition of BH's bytes into framing / explicit structure / residual. **Measurement:** explicit structure = **0–6% of the total** (3 images). The smart header is cheap; the expensive part was the residual (now delegated). *Answers the "killer question": making the structure explicit costs very little.*

Method (the union): document written as `.bh`, measuring the two faces in the same file.

Measurements: - Representation: **2.3 KB vs 4.8 KB for WebP = 2.1× smaller**. - Selective read: reading 1 region costs flat 87 B · gradient 96 B · text 590 B · diagram 1.8 KB — **3 to 55× less** than WebP, which needs the whole file for any region.

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WebP/AVIF → optimal residual, SINGLE read
PDF       → orchestrates specialists, but flat list, no selective read
OLAP/GPU  → selective read, but not a representation format
.bh       → BOTH faces + hierarchy + belonging, in a single envelope
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4. THE TRANSVERSAL LAW (measured across all angles)

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Hierarchy pays off when STRUCTURE dominates the cost, not dense SIGNAL.
At the limit: payload → PROGRAM that generates the data, to the extent that the data is
rule-generated. The boundary is not entropy — it is structure recognition.
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data class	measured result
procedural / formula	tiny program: 800–3,600× (decode-program)
compositional / symbolic	envelope 384× smaller + queries the dense one cannot do
heterogeneous structured	orchestration 2.1× + selective read 3–55×
aggregable / reused	selective read ∞–488× (database), 1,750× (GPU)
co-registered + redundant	wafer+temporal 2.13× (film)
dense / perceptual (photo, audio, embedding)	**loses or ties — connectionism's territory**

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5. THE REORGANIZED THESIS (supported by the evidence)

BH is not a codec — it is a structural orchestrator. Two faces, measured:

- **Read face** (summary in the node → reading selectively is cheap): GPU 1,750×, database ∞/488×, Merkle 52,000×, thumbnail/ROI 4–44×. Enormous numbers, but the mechanism is already SOTA.
- **Representation face** (explicit structure + delegated residual): orchestration 2.1× on a document. Smaller number, but **vs the state of the art**, by an architectural property.

The union of the two faces in one envelope (2.1× smaller AND navigable) is what no SOTA tool delivers. **The value is not in the compressed block — it is in the structure that knows what that block means.**

Three scales where BH survives scrutiny: 1. structural format (one asset) · 2. orchestrator (heterogeneous asset) · 3. compositional substrate (symbolic system). Common root: structure > signal.

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6. HONEST BOUNDARIES (so the thesis does not turn into faith)

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- "Wins for the right reason" ≠ "we found the product". It is scientific validation; the
  product requires construction and adoption (engineering, not benchmark).
- The orchestrator competes with PDF (the incumbent), not with WebP. Its differentiator
  — hierarchy + multiple reads — is real but unproven as a market need.
- The large numbers of the read face are vs NAIVE; vs SOTA, they tie. The
  differentiator is the UNION of the faces, not an isolated number.
- The decode-program requires RECOGNIZING the structure (inverse problem): trivial for
  a known family, undecidable in general.
- BH's home is STRUCTURE-DOMINANT data; in dense signal connectionism reigns,
  and BH DELEGATES to it — it does not confront it.
- A format dies without an ecosystem. The path is not a generic `.bh` competing
  with Parquet/PDF; it is building it native in a greenfield domain that already needs it.
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7. THE SELF-CORRECTIONS (evidence of rigor, not of weakness)

Three measurements were corrected in public when they proved to be biased:

1. **GPU — suboptimal flat kernel.** The raw number (4,725×) was inflated by a poor baseline implementation; the honest triangulated number is ~1,750×.
2. **Multimodal — embedding treated as irreducible payload.** Corrected with progressive representation (Matryoshka): storage moved from "loses" to "ties".
3. **Heterogeneous — hypothesis "wins on mixed content".** Refuted by the number (loses 10.75×); the real gain requires per-specialist routing (orchestration), not pure BH-compressor.

8. RELATED WORK (situating vs the state of the art)

BH does not invent the techniques it uses — it **unifies** them. Each isolated capability already exists and is mature; the contribution is to bring them together in a single structural envelope. We situate here the four families of prior art.

Adaptive per-block codecs. JPEG (Wallace, 1991) established the model "transform + quantization + entropy coding" with a fixed global basis (DCT). Modern codecs — AV1/AOMedia (Chen et al., 2018), VVC, JPEG XL — do *mode decision* per block, choosing prediction/transform modes per region. The BH-codec competes directly here and **loses** (§3.1): it lacks the entropy-coding machinery, and per-region selection is what AV1 already does. The difference of BH is not compression — it is delegating the region to the specialist (§3.9).

Selective-read structures. The "read the summary, not the data" read is the basis of OLAP/data cubes (Gray et al., 1997), zone-maps and per- row-group statistics in Apache Parquet, and *materialized views*. The gains of §3.2 and §3.5 (database, GPU) are exactly this mechanism — enormous against naive scan, but **already state of the art**. We recognize them as a credibility anchor, not novelty.

Authenticated structures. The verification face (§3.3) is the Merkle tree (Merkle, 1987), the basis of Git, Certificate Transparency (Laurie, 2014) and blockchains (Nakamoto, 2008). We do not invent crypto; we demonstrate that the Merkle-proof is the *same* read-by-objective-over-hierarchy as the other terrains.

Compact representation and data-as-program. For embeddings, the compressibility that §3.6/ §3.7 exploit is Product Quantization (Jégou et al., 2011) and Matryoshka Representation Learning (Kusupati et al., 2022); vector search is HNSW (Malkov & Yashunin, 2018). The "decode-program" of §3.8 is the direction of Kolmogorov complexity (Kolmogorov, 1965; Li & Vitányi) and has direct precedent in PostScript (page-as-program, Adobe) and procedural generation. Self-describing formats (HDF5, Protocol Buffers, PDF/ISO 32000) prove that "data that carries its own structure" works.

The gap that BH occupies. None of these unifies representation + selective read + specialist orchestration + hierarchy/belonging + proofs in a single envelope. PDF orchestrates but is a flat list without selective read; OLAP navigates but is not a representation format; AV1 adapts per block but with a single family of modes. BH's contribution is the **union**, not the components.

9. REPRODUCTION

```
codec      X:\bitH\           py -m pytest tests -q   ·  src/bench/{harness,headtohead,portfolio,
                               heterogeneous,cost_envelope,orchestrate,union}.py
database   X:\bitH\db\        py -m pytest tests -q   ·  src/bench/harness.py
merkle     X:\bitH\merkle\    py -m pytest tests -q   ·  src/bench/harness.py
wafer      X:\bitH\wafer\     py -m pytest tests -q   ·  src/bench/{harness,film}.py
gpu        X:\bitH\gpu\       py -m pytest tests -q   ·
src/bench/{real_gpu,heavy_gpu,extrapolate,native_model}.py
multimodal X:\bitH\multimodal\  src/{probe,fatia2,probe_progressive}.py
composit.  X:\bitH\compositional\
{probe_compositional,real_density,decode_program,program_test}.py
debt       X:\bitH\architecture_debt\  debt_decomposition.py

Python: X:/miniconda3/python.exe  (NumPy, Pillow, hashlib, CuPy/CUDA, sentence-transformers,
sklearn)
Raw reports per terrain: RESULTS_*.md in each folder.
```

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10. CONCLUSION

The study tested Hierarchical Bits from nine independent angles, with a declared method, correctness as a gate, honest baselines and public self-correction. The verdict:

> BH is not a better codec — it is a different way to **organize and read structured data**. Where the data is structure (rule-generated, composite, hierarchical, aggregable, co-registered), the gain is real and sometimes by orders of magnitude, and its deepest form is the payload becoming the *program* that generates the data. Where the data is dense signal (photo, audio, embedding), connectionism reigns, and BH's role is to **delegate** to the specialist — not to compete. The unique contribution is not an isolated number; it is the **union**, in a single envelope, of lean representation, selective read, hierarchy and belonging — which no current tool offers together.

The final proof is not one more benchmark; it is the construction of the format in a domain where structure and belonging matter as much as size.

From thesis to artifacts — three usable prototypes

The construction is no longer one step but three, and they share **one envelope** — proof the paradigm generalizes across domains. Each is a Python library with a measured demo and tests; each reads only the fraction a query asks for, in real bytes read from the file.

`bhmem` **— agent memory.** The agent writes structured memories (fact/event/relation + time + topic + provenance); the format exposes them through the multiple reads.

read	what it returns	bytes read	vs flat store (reads all)
`summary()`	summary of all topics	2.5%	**35× less**
`recall(topic)`	the memories of one branch	4.0%	**22× less**
`since(t)`	memories of the temporal window	9.8%	**9× less**
`provenance(id)`	source+path of 1 memory	10.8%	**8× less**

(Demo: ~90 days, 60 topics, 2,250 memories; 9/9 tests green.)

`bhtrace` — **distributed traces (observability).** A trace is a tree (trace → spans → events); the structure is the index, the heavy attributes (SQL, stack traces, headers) are the residual.

read	what it returns	vs flat store
`summary()` / `critical_path()`	the skeleton without the payload	**9× less**
`subtree(span)`	a drill-in	**9× less**
`service(name)`	one service	**5× less**

(Demo: 269-span checkout trace, attribute-heavy; 9/9 tests green.)

`bhckpt` — **model checkpoints.** A checkpoint is a hierarchy (model → layers → tensors → experts); the weights dominate, the structure is tiny.

read	what it returns	vs flat checkpoint
`summary()`	architecture of a 16 MB model read in 9 KB, no weights	**1,779× less**
`expert(i, e)`	load one MoE expert without the mixture	**20× less**
`layer(i)` / `tensor(name)`	one layer / one tensor	**12× / 10× less**

(Demo: 4-layer transformer, 2 MoE layers × 6 experts, 16.2 MB; 8/8 tests green.)

Honest boundary, the same across all three: BH wins on structural access and **delegates** the dense residual (full-text search → inverted index; per-tensor quantization → its specialist; dense semantic recall → a vector index). For `bhckpt`, selective per-tensor read already exists (`safetensors`) — that is the *anchor*; the new piece is the *union* (hierarchy including the MoE expert as a first-class read, per-tensor codec routing, a Merkle face for provenance).

These are not more measured angles — they are the thesis turning into tools, the **same envelope proven in three domains**. The gain scales with how structure-dominant the data is — the transversal law of §4. See the respective READMEs (`bhmem/` , `bhtrace/` , `bhckpt/`).

"The value is not in the compressed block. It is in the structure that knows what that block means."